॥ श्रीगणेशाय नमः ॥

# Mixtures and Alligation

Q. q1 kg of type-1 rice is sold at rs. p1/kg. q2 kg of type-2 rice is sold at rs. p2/kg. For what ratio q1:q2 do we get a mean price of rs. p/kg?

→

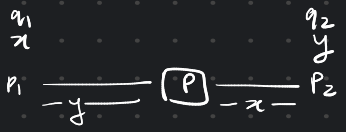
$$p = \frac{q_1 p_1 + q_2 p_2}{q_1 + q_2}$$

$$pq_1 + pq_2 = p_1 q_1 + p_2 q_2$$

$$q_1 (p - p_1) = q_2 (p_2 - p)$$

$$\frac{q_1}{q_2} = \frac{(p_2 - p)}{(p - p_1)}$$

If $q_1 : q_2 = x : y$




Q. The CP of 35 kg bananas is rs. 19.5 rs/kg. The CP of 30 kg bananas is rs. 22.10 kg. What's the avg price?

→

$$\frac{7}{6}\frac{35}{30} = \frac{221 - x}{x - 19.5}$$

$$7x - 136.5 = 132.6 - 6x$$

$$13x = 269.1$$

$$x = 20.7$$




$$19.5 + \frac{6}{13} \times 2.6$$

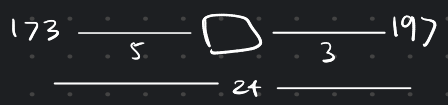
$$= 20.7$$

Q. 45 people. Avg height = 173 cm. Another 75 people. Avg height = 197 cm. Avg height in total?

→

45 : 75

3 : 5




$$173 + \frac{5}{8} \times \cancel{24}^{3}$$

$$= 173 + 15 = 188$$

or $197 - \frac{3}{8} \cdot 24$

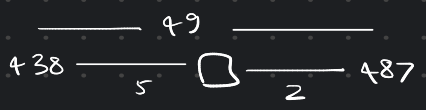
$$= 197 - 9 = 188$$

Q. 2 groups of students (38 and 95 students resp). Avg marks are 438 and 487 marks resp. Overall avg?

→

38 : 95

2 : 5




$$438 + \frac{5}{7} \times 49$$

$$438 + 35 = 473$$

## Removed and replaced

$$\frac{\text{final}_q}{\text{initial}_q} = \left(1 - \frac{x}{c}\right)^t$$

c = total capacity
x = how much removed and replaced
t = how many times you do it

Q. A vessel contains 80L of ethanol. 20L of this liquid is removed and replaced with water. 20L of this mixture is again removed and replaced with water. How much water (in L) is present in the vessel?

→ 80L Ethanol

↓

60L ethanol + 20L water

3 : 1
45L E 15L E
15L W 5L W

↓

60L 20L
3 : 1 mix water
45L ethanol 15L water + 20L water

45L ethanol + 35L water } Final mixture

35

Using formula:

Use formula if removed amount = replaced amount.

$$\frac{f_q}{i_q} = \left(1 - \frac{x}{c}\right)^t$$

$$\frac{f_q}{80} = \left(1 - \frac{20}{80}\right)^2$$

$$f_q = 80 \times \frac{60}{80} \times \frac{60}{80} = 45$$

= Final quantity of ethanol.

water = 80 − 45 = 35 L.

Q. A vessel contains 20L of acid. 4L acid is taken out and replaced with 4L water. 4L of this mixture is again replaced with water. Find the ratio of acid left in the vessel to the quantity of acid initially in the vessel.

→ $\frac{16}{20} \cdot \frac{16}{20} = \frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25}$

16 : 25

20L acid —4L repl.→ acid left = $\left(\frac{16}{20} \times 20\right)$ —4L repl.→ acid left = $\frac{16}{20} \times$ prev acid left

so acid = 20 − 4 = 16

total mixture = 20

$= \frac{16}{20} \times \left(\frac{16}{20} \times 20\right)$

Q. In a jar, 120L of milk is mixed with 24L of water. 12L of this mixture was taken out and 3L of water is added. If 27L of the newly formed mixture is taken out, what'll be the resultant quantity of water in it?

C can't use formula directly.

→ 120L milk 24L water

144L mixture
(5 : 1)
m w

↓

144 − 12 = 132 L

132L (5:1) 3L water

110L m 22L w 3L w

110L m 25L w

New mixture 110L m 25L w

135L mix.

↓

22 : 5

135 − 27 = 108

108 L
22 : 5

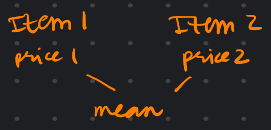
$w = \frac{5}{27} \times 108 = 20L$

Q. A vessel has 30L wine and 10L water. 4L mixture is removed and replaced with 4L water. This is repeated again. Ratio of wine left to initial wine is?

→ $\text{final}_{\text{wine}} = \underbrace{30}_{\text{initial}} \times \frac{36}{40} \times \frac{36}{40}$

$$\left(\frac{\text{final}}{\text{initial}}\right)_{\text{wine}} = \frac{36 \cdot 36}{40 \cdot 40} = \frac{9 \cdot 9}{10 \cdot 10} = \frac{81}{100}$$

# In what ratio should they be mixed?




ratio to be mixed in : $p_2$ − mean price : mean − $p_1$ price

Q. A gold vendor has 2 varieties of gold: 12 carets and 16 carets. In what ratio should he mix them to make an ornament of 15 carets purity?

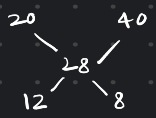
→ 16 − 15 : 15 − 12

1 : 3




Q. In what ratio should a 20% acid soln be mixed with a 40% acid soln to get a 28% acid soln?

→




3 : 2

Q. In 2 vessels A and B, there's a mixture of milk and water. The ratio of milk and water in this these vessels is 5:2 and 8:5 resp. In what ratio should these mixtures be mixed together so that the ratio of milk and water in the new mixture becomes 9:4?

→ consider milk




$\frac{5}{7} - \frac{9}{13}$

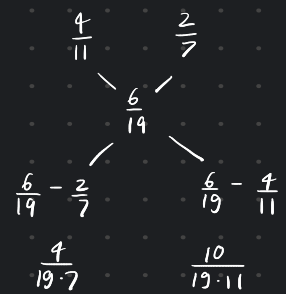
$\frac{65-63}{7 \cdot 13} = \frac{2}{7 \cdot 13}$

$\frac{\text{milk}_A}{\text{milk}_B} = \frac{1/13}{2/7 \times 13} = \frac{7}{2}$

$\Rightarrow \frac{\text{mixture}_A}{\text{mixture}_B} = \frac{7}{2}$ 7 : 2

Q. In 2 blends of mixed tea, the ratio of Darjeeling and Assam tea are 4:7 and 2:5 resp. In what ratio should these blends be mixed to get the ratio of Darjeeling and Assam tea as 6:13?

→

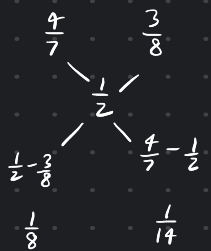



$\frac{2}{7} : \frac{5}{11}$

22 : 35

Q. 2 mixtures contain milk and water in ratios 4:3 and 3:5 resp. The ratio of the volumes if quantities from the vessels are to be mixed so that the new mixture has the same amounts of milk and water is?

→ new mixture → 1 : 1




14 : 8

7 : 4

## Mixing 3 mixtures (equal quantity)

Q. 3 equal glasses are filled with a mixture of milk and water in ratios 3:1, 5:3 and 9:7 resp. The contents of these are emptied into a single vessel. What's the ratio of milk and water in the new mixture?

→ $G_1$ $G_2$ $G_3$
3:1 5:3 9:7

milk

$G_1$ $G_2$ $G_3$

$\frac{3}{4} + \frac{5}{8} + \frac{9}{16}$

$\frac{12+10+9}{16}$

$\frac{31}{16}$

water

$G_1$ $G_2$ $G_3$

$\frac{1}{4} + \frac{3}{8} + \frac{7}{16}$

$\frac{4+6+7}{16}$

$\frac{17}{16}$

31 : 17

Q. 3 equal glasses contain milk and water in ratios 2:3, 3:4, and 4:5 resp. The contents of these are emptied into a single vessel.
What's the ratio of milk and water in the new mixture?

→

milk

$\frac{2}{5} + \frac{3}{7} + \frac{4}{9}$

$= \frac{401}{5 \cdot 7 \cdot 9}$ :

water

$\frac{3}{5} + \frac{4}{7} + \frac{5}{9}$

$\frac{544}{5 \cdot 7 \cdot 9}$

401 : 544

## Mixing 3 mixtures (unequal quantity)

Q. 3 glasses of volumes 3L, 4L, and 5L contain mixtures of milk and water in the ratios 2:3, 3:7, and 4:11 resp. The contents of all these are poured into a single vessel. Find the milk to water ratio in the resultant mixture.

→ milk :

$\frac{2}{5} \cdot 3 + \frac{3}{10} \cdot 4 + \frac{4}{15} \cdot 5$

$\frac{6}{5} + \frac{6}{5} + \frac{4}{3}$

$\frac{12}{5} + \frac{4}{3}$

water :

$\frac{3}{5} \cdot 3 + \frac{7}{10} \cdot 4 + \frac{11}{15} \cdot 5$

$\frac{9}{5} + \frac{14}{5} + \frac{11}{3}$

$\frac{23}{5} + \frac{11}{3}$

56 : 124
14 : 31

Q. 3 glasses of volumes in ratio 2:3:4 contain mixtures of milk and water in the ratios 4:1, 11:4, and 7:5 resp. The contents of all these are poured into a single vessel. Find the milk to water ratio in the resultant mixture.

→ milk :

$\frac{4}{5} \cdot 2 + \frac{11}{15} \cdot 3 + \frac{7}{12} \cdot 4$

$\frac{8}{5} + \frac{33}{15} + \frac{28}{12}$

water :

$\frac{1}{5} \cdot 2 + \frac{4}{15} \cdot 3 + \frac{7}{12} \cdot 4$

$\frac{2}{5} + \frac{12}{15} + \frac{28}{12}$

11 : 4

# Finding x

Q. A vessel contains milk and water in the ratio 1:x. When 300ml of the mixture and 50ml of water are mixed, the ratio of milk and water becomes 2:5. Find x.

→ 300 ml

$\frac{1}{1+x} \cdot 300$ milk  $\frac{x}{1+x} 300$ water

$$\frac{300}{1+x} : \frac{300x}{1+x} + 50 = 2:5$$

$$300 : 300x + 50 + 50x = 2:5$$

$$\frac{300}{350x+50} = \frac{2}{5}$$

$$\frac{6}{7x+1} = \frac{2}{5}$$

$$30 = 14x + 2$$

$$x = 2$$

Q. A vessel contains milk and water in the ratio 5:x. When 1L of water is added to 4L of the mixture, the resultant ratio of milk and water becomes 1:1. What's x?

→ 4L mixture

5 : x

$\frac{5 \cdot 4}{5+x}$ milk $\frac{4x}{5+x}$ water + 1L water

$= \frac{20}{5+x}$ milk + $\frac{5x+5}{5+x}$ water

$$\frac{20}{5x+5} = \frac{1}{1}$$

$$4 = x + 1$$

$$x = 3$$

Q. The students of classes x, y, z write a test and score avg marks 70, 81, and 90 resp. Avg of x and y is 76, of y and z is 86. Find avg of all 3 to the nearest integer.

→







# of students = 5 : 6

# of students : 4 : 5

# of students 10 : 12 : 15

$$\text{avg} = \frac{10(70)+12(81)+15(90)}{10+12+15} = \frac{3022}{37} = 81.67$$

$$\equiv 81$$

Q. A container has 1000L milk. 100L milk was taken out and replaced by water. This process was repeated 3 times.
How much milk is left?

Here it is done once. Then repeated 3 more times.

→ $1000 \cdot \frac{900}{1000} \cdot \frac{900}{1000} \cdot \frac{900}{1000} \cdot \frac{900}{1000} = 656.1$

Q. From a tank with 6000L mineral water, 150L water is replaced with normal water each time. If this is continued 4 times, how many ltrs of mineral water is left in the tank?

→ $6000 \times \frac{5850}{6000} \times \frac{5850}{6000} \times \frac{5850}{6000} \times \frac{5850}{6000} = 5422.1273$

This is done 4 times in total.

Q. 30L of a mixture contains 20% petrol and 80% kerosene. If 5 ltrs kerosene is added and 2 ltrs petrol is removed, what is ratio of petrol to kerosene?

→ 30L
- 6L petrol → 4
- 24L kerosene → 29

4 : 29